

Research Interests

Quantum Error Correction; Quantum Computer Architecture and Systems; Quantum Compilers

Education

2023.09 - **Carnegie Mellon University**

- Present ○ Ph.D. Student, Computer Science Department
- Advisor: Prof. Umut A. Acar

2019.09 - **Tsinghua University**

- 2023.06 ○ BEng in Computer Science, Institute for Interdisciplinary Information Sciences, founded by Prof. Andrew Yao
- Advisor: Prof. Xiongfeng Ma
- GPA: 3.98/4.00 (TOP 1/30)

Selected Honors and Awards

- 2023 **1st/73 Place** of ACM/IEEE Quantum Computing for Drug Discovery Challenge
- 2022-2023 **Honored graduate**, Beijing
- 2022-2023 **Honored graduate**, Tsinghua University (2/91)
- 2021-2022 **Comprehensive Excellent Award**, Tsinghua University
- 2021-2022 **Yao Award**, Institute for Interdisciplinary Information Sciences
- 2020-2021 **Comprehensive Excellent Award**, Tsinghua University
- 2020-2021 **Outstanding Youth**, Tsinghua University (2/91)
- 2019-2020 **Comprehensive Excellent Award**, Tsinghua University (1/91, Highest honor)
- 2018 **Gold Medal**, Chinese Physics Olympiad (Rank 7th nationwide)
- 2018 **First Prize**, Chinese Physics Olympiad Provincial (Rank 1st in Hebei Province)

Publications

*: Equal Contribution

- QCE'25 **ConiQ: Enabling Concatenated Quantum Error Correction on Neutral Atom Arrays**
Pengyu Liu, Mingkuan Xu, Hengyun Zhou, Hanrui Wang, Umut A. Acar, Yunong Shi
IEEE International Conference on Quantum Computing and Engineering
- QCE'25 **Local Optimization of Quantum Circuits**
Jatin Arora, Mingkuan Xu, Sam Westrick, Pengyu Liu, Dantong Li, Yongshan Ding, Umut A. Acar
IEEE International Conference on Quantum Computing and Engineering
Best Paper Award(1st place)
- SPAA'25 **POPQC: Parallel Optimization for Quantum Circuits**
Pengyu Liu, Jatin Arora, Mingkuan Xu, Umut A. Acar
ACM Symposium on Parallelism in Algorithms and Architectures
- QCE'24 **GraFeyn: Efficient Parallel Sparse Simulation of Quantum Circuits**
Sam Westrick, Pengyu Liu, Byeongjee Kang, Colin McDonald, Mike Rainey, Mingkuan Xu, Jatin Arora, Yongshan Ding, Umut A. Acar
IEEE International Conference on Quantum Computing and Engineering
Best Paper Award(2nd place)
- ISCA'24 **Atomique: A Quantum Compiler for Reconfigurable Neutral Atom Arrays**
Hanrui Wang, Pengyu Liu, Bochen Tan, Yilian Liu, David Z. Pan, Jason Cong, Umut Acar, Song Han
The International Symposium on Computer Architecture
- DAC'24 **Q-Pilot: Field Programmable Qubit Array Compilation with Flying Ancillas**
Hanrui Wang, Bochen Tan, Pengyu Liu, Yilian Liu, Jiaqi Gu, Jason Cong, Song Han
Design Automation Conference (DAC), 2024
- POPL'24 **SimuQ: A Domain-Specific Language For Quantum Simulation With Analog Compilation**
Yuxiang Peng, Jacob Young, Pengyu Liu, and Xiaodi Wu
51st ACM SIGPLAN Symposium on Principles of Programming Languages
Distinguished Paper Nominated

- PRL **Fundamental Limitation on the Detectability of Entanglement**
Pengyu Liu, Zhenhuan Liu, Shu Chen, and Xiongfeng Ma
Physical Review Letters 129, 230503 (2022)
Editors' Suggestion (Top 16%)
- TQC **Fundamental Limitation on the Detectability of Entanglement**
Pengyu Liu, Zhenhuan Liu, Shu Chen, and Xiongfeng Ma
Conference on the Theory of Quantum Computation, Communication and Cryptography
- PRL **Detecting Entanglement in Quantum Many-body Systems via Permutation Moments**
Zhenhuan Liu, Yifan Tang, Hao Dai, **Pengyu Liu**, Shu Chen, and Xiongfeng Ma
Physical Review Letters 129, 260501 (2022)
- ICCAD'22 **QuEst: Graph Transformer for Quantum Circuit Reliability Prediction**
Hanrui Wang, **Pengyu Liu**, Jinglei Cheng, Zhiding Liang, Jiaqi Gu, Zirui Li, Yongshan Ding, Weiwen Jiang, Yiyu Shi, Xuehai Qian, David Z. Pan, Frederic T. Chong, Song Han
IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2022
Workshop Papers
- APS March Meeting'26 **Efficient Atom Loss Decoding via Pauli Boundedness and Adaptive Re-weighting**
Pengyu Liu, Shi Jie Samuel Tan, Eric Huang, Umut A. Acar, Hengyun Zhou, Chen Zhao
- ScaleOPT at NeurIPS'25 **Quantum-Inspired Hamiltonian Descent for Mixed-Integer Quadratic Programming**
Shreya Chaudhary, Jinglei Cheng, Samuel Kushnir, Jiaqi Leng, **Pengyu Liu**, Yuxiang Peng, Hanrui Wang, Xiaodi Wu
- FastML at ICCAD'23 **ResilienQ: Boosting Fidelity of Quantum State Preparation via Noise-Aware Variational Training**
Hanrui Wang*, Yilian Liu*, **Pengyu Liu***, Song Han
- FastML at ICCAD'23 **TT-QEC: Transferable Transformer for Quantum Error Correction Code Decoding**
Hanrui Wang, **Pengyu Liu**, Kevin Shao, Dantong Li, Jiaqi Gu, David Z. Pan, Yongshan Ding, Song Han
Preprints & In-preparation
- Efficient Sampling and SAT-Based Verification of Magic State Cultivation Circuits**
Pengyu Liu, Steve Yueran Cao, Yunong Shi, Umut A. Acar
- ASPLOS'27 in Submission **QALM: Escaping Local Minima via Interleaved Exploration and Exploitation in Quantum Circuit Optimization**
Aidan Wagner*, Mingkuan Xu*, **Pengyu Liu***, Zhihao Jia, Umut A. Acar
- NeurIPS'27 in Submission **QIPrune: Quantum-Inspired Pruning of Large Language Models**
Zhiyuan Jia, **Pengyu Liu**, Yu-Hsuan Wu, Tabish Shaik, Jianhao Ma, Jiaqi Leng, Xiaodi Wu, Chaoyue Zhao, Yuxiang Peng
- PRX Quantum in Submission **Achieving Optimal-Distance Atom-Loss Correction via Pauli Envelope**
Pengyu Liu, Shi Jie Samuel Tan, Eric Huang, Umut A. Acar, Hengyun Zhou, Chen Zhao
- DGR: Tackling Drifted and Correlated Noise in Quantum Error Correction via Decoding Graph Re-weighting**
Hanrui Wang*, **Pengyu Liu***, Yilian Liu, Jiaqi Gu, Jonathan Baker, Frederic T. Chong, Song Han

Teaching Experiences

- 2026 **Teaching Assistant** "15-898: Quantum Computing Systems"
- 2026 **Teaching Assistant** "15-459: Undergraduate Quantum Computation"
- 2023 **Teaching Assistant** "General Physics"
- 2022 **Teaching Assistant** "Quantum Communication and Cryptography"

Working and Exchanging Experiences

- 2026.05 - **Amazon (AWS Center for Quantum Computing)**, Pasadena, CA, Applied Scientist Intern
- 2026.08
- Advised by Dr. Catherine Leroux.
 - Developed belief-propagation decoding methods for qLDPC codes under biased noise models.
- 2025.05 - **QuEra Computing Incorporated**, Boston, MA, Research Intern
- 2025.08
- Advised by Dr. Chen Zhao.
 - Designed simulators for atom-loss effects in quantum error correction, characterized their impact, and developed syndrome extraction and decoding strategies.
- 2022.07 - **Massachusetts Institute of Technology**, Boston, MA, Research Assistant
- 2023.06
- Advised by Prof. Song Han.
 - Worked on quantum machine learning and quantum computer architecture.

2022.03 - **University of Maryland, College Park, MD**, Research Assistant

- 2022.11
 - Advised by Prof. Xiaodi Wu.
 - Worked on analog quantum simulation.

Academic Services

- 2026 Organizing Committee of [Quantum Computing Applications and Systems Workshop \(QCAS'26\)](#)
- 2025-present Organizing Committee of [Quantum Computer System Lecture Series](#)
- 2022-2023 Organizing committee of [Yao Seminar](#)

Journal Reviewer,

- IEEE Transactions on Quantum Engineering
- ACM Transactions on Quantum Computing
- Journal of Parallel and Distributed Computing
- Quantum Information Processing
- Future Generation Computer Systems
- ACM Computing Surveys

Conference Reviewer,

- IEEE International Conference on Quantum Computing and Engineering (2024, 2025, 2026),
- ACM Symposium on Parallelism in Algorithms and Architectures (2025)